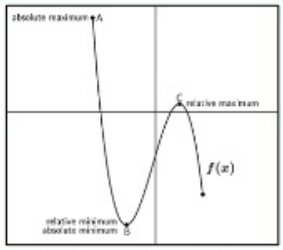
FUN-1.C.1 If a function f is continuous over the interval (a, b), then the Extreme Value Theorem guarantees that f has at least one minimum value and at least one maximum value on [a, b].

FUN-4.A.3 Absolute (global) extrema of a function on a closed interval can only occur at critical points or at endpoints.

5-2 & 5-5 Extreme Value Theorem and Absolute Extrema

We have previously discussed **Relative (local) Extrema** and the 1st derivative test. Recall a relative extremum may or may not be the MOST extreme value of the function over a given interval. The greatest value of a function $f(x)$ is called the **ABSOLUTE (Global) Maximum** and the least value of a function $f(x)$ is called the **ABSOLUTE (Global) Minimum**.



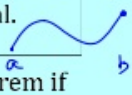
The function f is shown in the graph is on a closed interval.

The graph of f has a relative minimum at Point B and a relative maximum at point C.

The graph of f has its absolute minimum at Point B and its absolute maximum at Point A.

Important Note: Both types of extrema (relative and absolute) are important in Calculus. Since both are types of extrema, we might expect that the processes to find them will have some similarities. But it is important to note that (generally) we cannot use the 1st Derivative Test to find or justify an absolute extremum.

Extreme Value Theorem (EVT)	If f is **continuous** on the **closed interval** $[a, b]$, then f has both an absolute minimum value and an absolute maximum value in the interval.



Every theorem has **conditions**. We can only apply the theorem if these conditions are met!!!

If conditions are not met, then we cannot apply the theorem and must consider another approach (if possible). For the EVT, what are the condition(s)?

f is continuous on a closed interval

If the condition(s) are met, what are we guaranteed?

f has both an absolute minimum value and an absolute maximum on the closed interval.

Remember, this is what we call an **existence** theorem. The other two existence theorems we have learned in AP Calculus are __MVT_____ and __IVT_____.

While EVT guarantees that any function that meets the conditions has both an absolute max and absolute min, the theorem does not give us any information on HOW to actually find those values.

Candidates Test for Absolute (Global) Extrema

1. Find the critical values in the given interval.
2. Create a table of values—include the **critical values** and **endpoints**.
3. When finding absolute extrema—plug x values into the **function equation**!
4. The greatest value is the absolute maximum, and the least value is the absolute minimum.
5. Give the absolute minimum (and/or absolute maximum) using the following wording:

   The absolute minimum value on _____ is _____ at $x =$_____

   The absolute maximum value on _____ is _____ at $x =$_____

Note: The following statements are all equivalent ...
What is the absolute maximum of $f(x)$?
What is the global maximum of $f(x)$?
Find the maximum value of $f(x)$.
What is the maximum of $f(x)$?

Example 1: Find the absolute extrema for the following functions.

a. $f(x) = 2x^2 - 8x + 1$ on $[0, 3]$

$f'(x) = 4x - 8$

$f'(x) = 0$ when

$4x - 8 = 0$

$4x = 8$

$x = 2$

x	$f(x)$	
2	$8 - 16 + 1 = -7$	← min
0	1	← max
3	$18 - 24 + 1 = -5$	

the abs. max value on $[0,3]$ is 1 at $x = 0$

the abs. min value on $[0,3]$ is -7 at $x = 2$

Example 1: Find the absolute extrema for the following functions.

c. $h(x) = \dfrac{x^2}{x+3}$ on $[-2, 1]$

$$h'(x) = \frac{(x+3)2x - x^2(1)}{(x+3)^2}$$
$$= \frac{2x^2+6x-x^2}{(x+3)^2}$$
$$= \frac{x^2+6x}{(x+3)^2}$$

x	$h(x)$	
0	0	← min
-2	4	← max
1	$\frac{1}{4}$	

$h'(x)=0$ when
$x^2+6x=0$
$x(x+6)=0$
$x=0 \quad x=-6$

$h'(x)$ is undef when
$(x+3)^2=0$
$x+3=0$
$\cancel{x=-3}$ not a critical

the abs max value on $[-2,1]$ is 4 at $x=-2$
the abs min value on $[-2,1]$ is 0 at $x=0$

Example 1: Find the absolute extrema for the following functions.

d. $y = 2\sin(x) - x$ on $[0, 2\pi]$

$y' = 2\cos x - 1$

$y' = 0$

$2\cos x - 1 = 0$

$2\cos x = 1$

$\cos x = \frac{1}{2}$

$x = \frac{\pi}{3}$

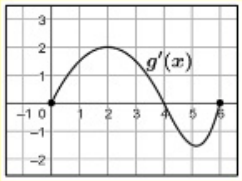
$x = \frac{5\pi}{3}$

x	y	
0	0	
2π	-2π	≈ -6.28
$\frac{\pi}{3}$	$\sqrt{3} - \frac{\pi}{3}$	≈ 0.685 max
$\frac{5\pi}{3}$	$-\sqrt{3} - \frac{5\pi}{3}$	≈ -6.968 min

the abs max value is $\sqrt{3} - \frac{\pi}{3}$ at $x = \frac{\pi}{3}$ on $[0, 2\pi]$

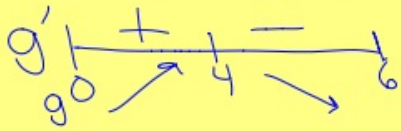
the abs min value is $-\sqrt{3} - \frac{5\pi}{3}$ at $x = \frac{5\pi}{3}$ on $[0, 2\pi]$

Example 4: The function g is differentiable on the closed interval $[0, 6]$. The graph of g', the derivative of g, is given above. At which x value does the graph of g have its absolute maximum value?



Critical values of g $[0,6]$:

$x=0$, $x=4$, $x=6$



g has its abs max value on $[0,6]$ at $x=4$ because $g'(x) \geq 0$ (g is inc) on $[0,4]$ and $g'(x) \leq 0$ (g is dec) on $[4,6]$

Example 1: Find the absolute extrema for the following functions.

b. $g(x) = \dfrac{(x^2 - 1)^2}{2}$ on $[-3, 2]$